

SPPL-374: Min Max DoH Post API (Apply To All Groups & Months True Only) technical workflow

Jira Ticket: [SPPL-374: BE | Create technical workflow for Min Max DoH Post API \(Apply to all groups & all months only\) on parameter inputs page including DB details](#) DONE

- i** This page provides comprehensive information about the POST API for Min Max DoH on the Input params page, to update Min Max DoH values for the scenario by groups.

Supply Planning POST Min Max DoH API Details:

1. **Hosted URL:** <https://api.spln.cube<env>.toyota.com/<env>>
2. **API Name:** <HOSTED_URL>/v/1/1/sp/min-max-doh
3. **Path Param:** N/A
4. **Query Param:** N/A
5. **Method:** POST
6. **Authorization:** Bearer token (**required**)
7. **Lambda Name:** spln-<env>-update-min-max-doh-v1-1
8. **API Request Body:**

```
1 {  
2   "scenarioId": "uniqueScenarioId",  
3   "userEmail": "user@toyota.com",  
4   "applyTo": "ALL_GROUPS",  
5   "mode": "ALL_MONTHS",  
6   "data": {"min": 14, "max": 10 }  
7 }
```

9. API Response:

- o **Success response with status 200:**

```
1 {
```

```
2  "message": "Successfully updated data."
3  }
```

- **Error Response:**

- **any internal server error occurred with status 500:**

```
1  {
2    "errorMessage": "Internal Server Error"
3  }
```

- **any unauthorized error occurred with status 401 (Managed by the AWS API Gateway integrated Custom Authorizer):**

```
1  {
2    "message": "Unauthorized"
3  }
```

- **If request is in-valid with status 400:**

```
1  {
2    "errorMessage": [<"ValidationError: validation error message">]
3  }
```

- **List of possible validation error message:**

1. "ValidationError: Request body cannot be empty."
2. "ValidationError: scenarioId is required and must be a uuid."
3. "ValidationError: userEmail is required and must be a string."
4. "ValidationError: Invalid userEmail."
5. "ValidationError: Scenario doesn't exist."
6. "ValidationError: applyTo is required and must be either ALL_GROUPS or BY_GROUPS."
7. "ValidationError: mode is required and must be either ALL_MONTHS or BY_MONTHS."
8. "ValidationError: data is required and must be an object."
9. "ValidationError: min is required and must be a non-negative number between 0 to 100."
10. "ValidationError: max is required and must be a non-negative number between 0 to 100."
11. "ValidationError: max must be greater than min."

10. **Table:**

- a. **scenarios**
- b. **grouping**
- c. **group_scenario_mapper**

d. **min_max_doh**

e. **scenario_step_status**

SP Update Grouping Data API Workflow:

• Technical workflow:

◦ Description:

- i. The Supply Planning user logs in to the application.
- ii. After the user provides creates a scenario, opens it, navigates to Min Max DoH and updates group days on hand data, a request is sent to the SP Post Min Max DoH endpoint: **sp/min-max-doh** with payload:

```
1 {  
2   "scenarioId": "uniqueScenarioId",  
3   "userEmail": "user@toyota.com",  
4   "applyTo": "ALL_GROUPS",  
5   "mode": "ALL_MONTHS",  
6   "data": {"min": 14, "max": 10 }  
7 }
```

- iii. After successful authentication, the **spln-update-min-max-doh-v1-1** lambda is triggered by the SP update scenario Min Max DoH endpoint: **sp/min-max-doh**. If authentication fails, an error with status 401 (*refer: error response-**any unauthorized error occurred with status 401***) will be returned and sent to client.
- iv. After user authentication has been confirmed, the **spln-update-min-max-doh-v1-1** lambda will perform validation on the request payload. If payload is valid will proceed to next step else throw validation error (*refer: error response: **Section if request is in-valid with status 400 and List of possible validation error message***). Below query is used to check if provided scenario is valid:

```
1 select * from supply_planning.scenarios where scenario_id = :scenarioId and  
   is_active = true;
```

- v. After the request payload is successfully validated, the **spln-update-min-max-doh-v1-1** lambda execute below logic:
 1. As the input is for “ALL_MONTHS”, get scenario timeframe from query (iv) result.
 2. As the input is for “ALL_GROUPS”, get active groupIds for scenario (from mapper) using below query:

```
1 SELECT sgm.group_id AS "groupId"  
2 FROM supply_planning.group_scenario_mapper sgm  
3 JOIN supply_planning.grouping gp  
4   ON gp.group_id = sgm.group_id  
5 WHERE sgm.scenario_id = ${scenarioId}::uuid  
6       AND sgm.is_active = TRUE  
7       AND gp.is_active = TRUE  
8 ORDER BY sgm.group_id
```

3. Extract months & years data from scenario timeframe from step 1 using below function

```
1  /**
2   * Build a list of {year, month} between startDate and endDate (inclusive),
3   * month granularity (month is 1-12).
4   * @param {Date} startDate
5   * @param {Date} endDate
6   * @returns {{year:number, month:number}[]}
7   */
8  function buildScenarioMonths(startDate, endDate) {
9      const monthsYears = [];
10
11     // normalize to first day of month
12     const cursor = new Date(Date.UTC(startDate.getUTCFullYear(),
13     startDate.getUTCMonth(), 1));
14     const end = new Date(Date.UTC(endDate.getUTCFullYear(),
15     endDate.getUTCMonth(), 1));
16
17     while (cursor <= end) {
18         monthsYears.push({
19             year: cursor.getUTCFullYear(),
20             month: cursor.getUTCMonth() + 1
21         });
22         cursor.setUTCMonth(cursor.getUTCMonth() + 1);
23     }
24
25     return monthsYears;
26 }
```

4. Bulk UPSERT query with apply_to_all_months true — upsert min_max_doh table data by scenarioId, groupId, month and year using below query:

```
1  // Build VALUES list with bound parameters.
2  // Each row expands to: (group_id, scenario_id, month, year, min_doh,
3  // max_doh, apply_to_all_months, created_by)
4  const valuesSql = Prisma.join(
5      rows.map(r => Prisma.sql`
6          ${r.groupId}::uuid,
7          ${scenarioId}::uuid,
8          ${r.month}::int,
9          ${r.year}::int,
10         ${minDoh}::int,
11         ${maxDoh}::int,
12         ${applyToAllMonths}::boolean,
13         ${userEmail}::text
14     `),
15      Prisma.sql`, `
16  );
17
18  const upsertSql = Prisma.sql`
19      INSERT INTO supply_planning.min_max_doh (
20          group_id,
21          scenario_id,
22          month,
23          year,
24          min_doh,
25          max_doh,
26          apply_to_all_months,
27          created_by
28      )
29      VALUES ${valuesSql}
30      ON CONFLICT (scenario_id, group_id, month, year)
31      DO UPDATE SET
```

```

31     min_doh = ${minDoh}::int,
32     max_doh = ${maxDoh}::int,
33     apply_to_all_months = ${applyToAllMonths}::boolean,
34     updated_by = ${userEmail}::text,
35     last_updated_timestamp = CURRENT_TIMESTAMP
36 `;

```

5. Update the step & scenario status to “In Progress”. Only update scenario status to “In Progress” if its not already using scenarioData from query (iv) response.

```

1  insert into supply_planning.scenario_step_status (scenario_id, input_step_type,
2  input_step_name, status, updated_by, last_updated_timestamp)
3  values (${scenarioId}::uuid, ${stepType}, ${stepName}, ${status}, ${userEmail},
4  now())
5  on conflict (scenario_id, input_step_name)
6  do update set status = ${status}, updated_by = ${userEmail},
7  last_updated_timestamp = now()
8  returning scenario_step_status_id::uuid;
9
10 UPDATE supply_planning.scenarios
11 SET scenario_status = "In Progress",
12    last_updated = CURRENT_TIMESTAMP
13 WHERE scenario_id = ${scenarioId}::uuid
14 returning scenario_id;

```

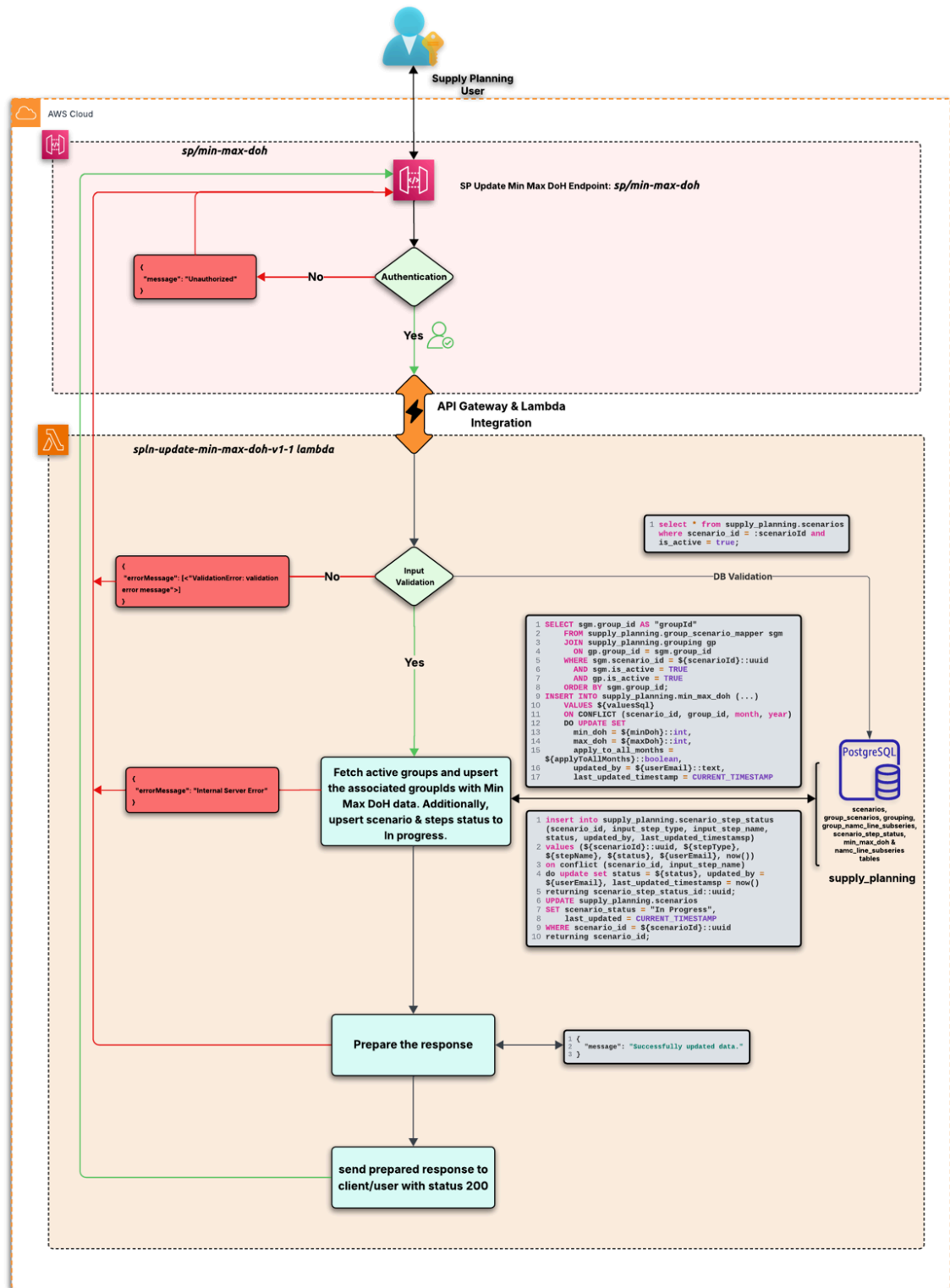
- vi. After updating the Min Max DoH & status data, return the below response to the client with a status code of 200:

```

1  {
2    "message": "Successfully updated data."
3  }

```

- vii. If there is an error while fetching the data, an “Internal Server Error” with a status code of 500(refer: error response-**any internal server error occurred with status 500**) will be returned. Otherwise, a successful response with a status code of 200 will be sent.



SP Min Max DoH Post API Technical workflow

Note:

We will be utilizing this API for both to update group data and create new groups.

DB constraints:

```
ALTER TABLE supply_planning.min_max_doh  
ADD CONSTRAINT uq_min_max_doh_scenario_group_month_year  
UNIQUE (scenario_id, group_id, month, year);
```

? Queries:

1. What is the maximum number of groups that can be linked to a scenario? 15
2. Max DoH value must be less than?
3. Min DoH value must be less than? @Daniel Diago